

Section 4 :The Future of Unmanned Aerial Vehicles

- Drones will provide solutions to solve many problems by increasing both flight time and load carrying capacity.
- The most common drone power is lithium batteries, and this works well with lightweight wing-type drones or quadcopters when doing short distances.
- The lithium batteries are not suitable for heavy loads and even the batteries that are used for electric cars still won't work for the heavy loads.
- The possible power source for drones to carry heavy loads are hybrid electric gasoline engines (powered by an internal combustion engine and one or more electric motors, which uses energy stored in batteries) and hydrogen engines (Hydrogen internal combustion engine vehicles are different from hydrogen fuel cell vehicles (which use electrochemical use of hydrogen rather than combustion)).
- Drones are getting bigger and stronger and soon will revolutionize cargo deliveries such as carrying large payload which will be able to reach destination that helicopters, and light vehicles find hard to reach.
- Unmanned delivery service is just beginning to take off such as Alphabet's Wing drones can carry 2.5pounds (1.1kg).
- Today prototypes such as Rhaegal drone lift 829pounds (376Kg) but this specific drone is equipped with a hybrid engine that converts aviation fuel energy into electricity. This certain service will be in demand in the field of construction and maintenance of complex in repairing power transmission towers etc.
- The future of Drones is driving in the direction of carrying passengers.
- Drones will get smarter than AI (Artificial Intelligence) by being operated by AI with no operator to fly along given routes and avoid collisions with other drones, buildings, and natural obstacles. Drones will be able to make their own decisions based on information they have analyse.
- The next development will be swarming of drones by uses of different groups will be capable of using one or more units because of the demand exclusively by army units for reconnaissance and combat operations.
- As for the business sector drones will be service and this service will be called Drone-as-a-Service (Daas). Daas will assist with the organizations to hire drones to carry out mapping study of a large area, monitor crops or livestock, or set up temporary mobile networks using flying towers (Drones that transmit a signal.).

6 ways Drones will shape our Future.

1. **Delivering Packages** : UAV are being used as a delivery services which means reducing the number of delivery vehicles and costs, faster delivery, lower pollution and more. This means that you will have more efficient deliveries, better quality air, and more to time to carry out more activities.
2. **Drone in the Industrial Inspections** : Construction companies, insurance, building engineers and architects, are already doing business envelope inspections using drones. Drones are also used in the Oil and Gas sector to perform inspections and collecting data quickly and this will reduce cost at the same time reducing the risk of rope access, swing

stages and scaffoldings. This is very essential because employees of the Oil and Gas sector are three times more likely to be involved in accidents compared to other employees of the other sector.

3. **Drones in Agriculture:** Drones in the agriculture sector will assist in detecting infested crops, patches or group of insects on crops or field. The UAV with the right equipment will be able to spray the affected crops if not it will be able to send information, location and the type of bug to another drone equipped with the right spray to spray the bug. It may even send information to the central control room for action.
4. **UAVs in Filming:** Drones are a great marketing tool they are able to assist in captivating content and video campaigns. Drones have the advantage of capturing high-quality image and video in a short time. Drones have the ability to take photos from any angle and all directions. Real estate firms, tourist resorts, school motion picture, and photography etc will have the advantage of the drones for their clients.
5. **Transporting Supplies For Humanitarian Work:** In future drones will transport supplies to remote or inaccessible areas. Drones will deliver medical supplies and vaccines to remote locations, dangerous places, inaccessible places as well as places affected by war and contagious places. As for the other application it will assist in rescue missions, disaster management, healthcare and more which will increase the safety of rescue workers.
6. **Expand Internet Access:** Solar powered drones will have the ability to expand internet access to remote areas. A network of drones with internet links will beam the internet connectivity to users in remote areas.

The Following Cases that show the improvement of Unmanned Aerial vehicles

1. South African National Parks (SANParks) is using drones for wildlife conservation. The drones are equipped with thermal cameras and GPS tracking systems to monitor and protect endangered species, particularly rhinos, from poaching.
2. The mining company Anglo American is using drones to inspect their mines in South Africa. The drones are equipped with high-resolution cameras and laser scanners to detect potential safety hazards and map underground structures.
3. The humanitarian organization Gift of the Givers has used drones to deliver food, water, and medical supplies to remote areas affected by drought and flooding. The drones have been particularly useful in areas that are difficult to access by road.
4. The agricultural technology company Aerobotics is using drones to help farmers monitor their crops. The drones take high-resolution images of fields, which are analyzed by artificial intelligence algorithms to detect potential issues such as pest infestations or nutrient deficiencies.

5. The security company SkyEye is using drones to monitor high-crime areas in South Africa. The drones are equipped with high-resolution cameras and real-time streaming capabilities, allowing law enforcement agencies to respond quickly to potential threats.
6. The infrastructure inspection company WSP is using drones to inspect bridges, highways, and other structures. The drones are equipped with cameras and sensors that can detect potential issues such as cracks or corrosion, allowing for timely maintenance and repairs